

Mid-Semester Exam – Learning
(Total Marks: 5)

Q1: A pet trainer taught a dog to run towards an object when the word “Fetch” is called. He used classical conditioning for the training. Identify the UCS, CS, UCR, CR. (2 marks)

Q2. Ravi, a 10-year-old boy, had to visit the hospital for an injection, an experience he found extremely painful and frightening. The doctor who administered the shot was wearing a white coat, and ever since that day, Ravi felt uneasy whenever he saw someone dressed the same way. On subsequent visits, just seeing the hospital building from a distance made his heart race, even before stepping inside. Over time, as he waited in the reception area during check-ups, he remained calm till the sound of the receptionist’s voice calling out names started making him anxious, though he didn’t know why. Using your knowledge of higher order conditioning, identify the chain of conditioning by mentioning all the UCS and CSs. (3 Marks)

Answer Key & Marking Scheme

Q1: Training a Dog Using Conditioning (2 Marks)

Unconditioned Stimulus (UCS): Throwing the ball (0.25 marks)

Unconditioned Response (UCR): Dog instinctively runs after the ball (1 mark)

Conditioned Stimulus (CS): The command “Fetch” (0.25 marks)

Conditioned Response (CR): Dog runs after the ball upon hearing the command (0.5)

Q2: Higher Order Conditioning:

Pain due to Injection (UCS) -> Injection (CS_1) -> White Coat (CS_2) -> Receptionist's voice (CS_3) -> Hospital Building (CS_4) (0.5 marks for each = $0.5 \times 5 = 2.5$)

0.5 marks for identifying the correct sequence.

Q3: Difference between classical and operant conditioning (2 Marks)

Classical Conditioning	Operant conditioning
Behavior is elicited (automatic)	Behavior is emitted (voluntary)

Behavior need not be learned, it is naturally present	Response can be new behaviors, not naturally present
Behavior is dependent on Autonomic Nervous System	Behavior is dependent on skeletal muscles system
S-S (example)	S-R (example)

Q4:

Type of learning: Classical Conditioning.

Key elements:

Unconditioned Stimulus (US): Loud beeping sound (naturally causes annoyance and grogginess).

Unconditioned Response (UR): Feeling annoyed and groggy (automatic response to the beeping).

Conditioned Stimulus (CS): Alarm clock (initially neutral but paired with the beeping).

Conditioned Response (CR): Feeling annoyed and uneasy (learned response to seeing the alarm clock).